

Superluminal Travel in the Triadic Cosmos

Technical Impossibility, Causality Violation, Empirical Non-Realisation

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Abstract

Superluminal travel has long been imagined as a hallmark of advanced Kardashev civilizations, yet within the Triadic Cosmos framework it proves technically impossible, causally inconsistent, and empirically unsupported. This paper develops a unified no-go argument across five layers. First, we examine the extreme hypothetical of a “Dyson Spaceship”—a stellar-scale propulsion system enclosed in a Dyson sphere and driving a superluminal warp bubble. Because the universe is saturated with electromagnetic radiation and gravitational-wave backgrounds, such an object would produce unmistakable observational signatures, including superluminosity zones and spacetime compression artefacts, none of which have ever been detected. Second, even if superluminal transport were technically feasible, it would behave as an unguided projectile: a faster-than-light bullet traversing spacetime without external sensing, feedback, routing, or navigational correction. Third, we show that all known superluminal mechanisms—warp bubbles, wormholes, Krasnikov tubes, tachyonic channels—require exotic matter, negative-energy configurations, or closure-breaking physics that are explicitly excluded by triadic filtering and local-closure constraints. Fourth, superluminal motion violates causality in at least four independent ways: holographic bulk–boundary consistency, electromagnetic information propagation, gravitational-wave update latency, and classical cause–effect ordering, each of which breaks the Universal Closure–Boundary Normalisation Principle. Finally, we integrate these empirical, technical, and causal failures into a single comprehensive no-go theorem, demonstrating that superluminal travel is structurally impossible not only within the Triadic Cosmos but within any universe obeying consistent closure, unitarity, and causal propagation. The result is a broad, accessible, and conceptually unified refutation of faster-than-light transport.

1 Introduction

Superluminal travel occupies a peculiar position in both scientific speculation and science-fiction narratives. It is often imagined as the defining capability of advanced civilizations, a technological threshold beyond relativistic constraints and a hallmark of Kardashev Type II or higher societies. Within the Triadic Cosmos ecosystem, however, faster-than-light (FTL) transport is not merely technologically challenging: it is structurally incompatible with the foundational principles governing closure, causality, and boundary normalisation. This paper develops a unified and accessible no-go argument showing that superluminal travel cannot exist in any universe that respects consistent causal propagation, holographic completeness, and triadically filtered physical ontology.

We begin with an empirical perspective. If superluminal transport were possible, especially at scales relevant to hypothetical megastructures, its signatures would be unmistakable. To illustrate this, we introduce the extreme conceptual scenario of a “Dyson Spaceship”—a stellar-scale propulsion system enclosed within a Dyson sphere and driving a superluminal warp bubble. Because the universe is saturated with electromagnetic radiation and gravitational-wave backgrounds, such an object would generate

observable compression artefacts, superluminosity zones, and spacetime distortions detectable across cosmological distances. No such phenomena have ever been observed, and even smaller warp-like structures would produce detectable anomalies that are likewise absent.

Next, we examine the navigational and dynamical implications of superluminal motion. Even if a warp mechanism were technically realisable, it would behave as an unguided projectile: a faster-than-light bullet traversing spacetime without external sensing, feedback, routing, or correction. The causal horizon surrounding any superluminal bubble prevents information exchange with the external universe, rendering navigation impossible and reducing the vehicle to blind propagation through an unknown environment.

We then address the technical constraints. All known proposals for superluminal transport—including warp bubbles, wormholes, Krasnikov tubes, and tachyonic channels—require exotic matter, negative-energy configurations, or closure-breaking physics that violate the triadic filtering rules and local-closure constraints established within the Triadic Cosmos. These mechanisms cannot be embedded in a consistent triadic ontology and fail to satisfy the structural requirements for stable physical realisation.

Finally, we analyse the causal implications. Superluminal motion violates causality in multiple independent ways: holographic bulk–boundary consistency, electromagnetic information propagation, gravitational-wave update latency, and classical cause–effect ordering. Each violation independently breaks the Universal Closure–Boundary Normalisation Principle, and together they demonstrate that FTL transport is incompatible with any universe that maintains coherent causal structure.

By integrating these empirical, technical, and causal arguments, we formulate a comprehensive no-go theorem showing that superluminal travel is structurally impossible not only within the Triadic Cosmos but within any causally consistent physical framework. The result is a broad, conceptually unified, and accessible refutation of faster-than-light transport, grounded in the triadic principles that govern closure, boundary, and causality across the ecosystem.

2 Empirical Constraints: The Absence of Superluminal Signatures

Any physically meaningful analysis of superluminal transport must begin with empirical considerations. If faster-than-light motion were possible at scales relevant to advanced civilizations, its observational footprint would be unmistakable. The universe is permeated by electromagnetic radiation, cosmic microwave background photons, high-energy particle fluxes, and a continuous spectrum of gravitational waves. A superluminal object moving through this medium would necessarily interact with these fields in ways that produce detectable anomalies. In this section, we examine these implications through both extreme and modest hypothetical scenarios.

2.1 The Dyson Spaceship Hypothesis

To illustrate the empirical stakes, we introduce the conceptual scenario of a “Dyson Spaceship”—a Kardashev Type II propulsion architecture in which a star is enclosed within a Dyson sphere and used as the energetic driver of a superluminal warp bubble. Although entirely speculative, this scenario serves as a useful upper bound: if superluminal travel were feasible, a stellar-scale engine would represent the most powerful and technologically unconstrained implementation.

A warp bubble of such magnitude would compress the surrounding electromagnetic field, producing superluminosity zones, directional radiation spikes, and spacetime shear patterns detectable across intergalactic distances. Gravitational-wave backgrounds would likewise be distorted, as the bubble’s superluminal motion would outpace the propagation of curvature updates, leaving behind a characteristic trail of delayed spacetime responses. No such signatures have ever been observed in any astronomical survey, including those capable of detecting far subtler anomalies.

2.2 Absence of Smaller-Scale Warp Artefacts

Even if stellar-scale propulsion systems are dismissed as extreme, smaller warp-like structures would still produce detectable artefacts. Any superluminal bubble, regardless of size, would generate:

- compression of ambient electromagnetic radiation,
- directional brightening or dimming consistent with spacetime shear,
- gravitational-wave discontinuities or latency trails,
- anomalous motion inconsistent with relativistic kinematics.

Modern astronomical instruments—including wide-field surveys, gravitational-wave detectors, and high-resolution interferometers—have not detected any phenomena consistent with these predictions. The empirical absence of such artefacts places strong observational constraints on the existence of superluminal transport, independent of any theoretical or causal considerations.

2.3 Empirical Non-Realisation as a Structural Indicator

Within the Triadic Cosmos framework, empirical non-realisation is not merely a lack of evidence; it is a structural indicator. The ecosystem’s predictive architecture asserts that any closure-breaking phenomenon of sufficient magnitude must produce observable distortions in the triadic layers of electromagnetic, gravitational, and causal propagation. The absence of such distortions therefore constitutes positive evidence that superluminal travel does not occur, even at the highest conceivable technological levels.

Empirical constraints thus form the first pillar of the broader no-go theorem developed in this paper: if superluminal travel were possible, the universe would reveal its presence. It does not.

3 The Unguided Projectile Argument: Superluminal Motion as Blind Propagation

Even if superluminal transport were technically realisable, its operational behaviour would be fundamentally incompatible with controlled navigation. A faster-than-light vehicle cannot interact with its external environment while in motion, because any superluminal bubble necessarily forms a causal horizon that prevents information exchange between the interior and the surrounding spacetime. This horizon is not a technological limitation but a structural consequence of superluminal propagation itself. As a result, any object traveling faster than light becomes an unguided projectile: a blind, uncorrectable trajectory through an environment it cannot sense.

3.1 Causal Isolation of the Warp Interior

The interior of a superluminal bubble is causally disconnected from the external universe. No electromagnetic signals, gravitational perturbations, or material interactions can cross the bubble boundary in either direction. This implies that:

- external sensors cannot provide situational awareness,
- internal instruments cannot detect obstacles or hazards,
- no feedback loops can stabilise or adjust the bubble,
- no routing or course correction is possible,
- no emergency shutdown can be triggered from outside.

The vehicle is therefore unable to respond to its environment. It cannot detect incoming matter, radiation, or gravitational structures, nor can it avoid collisions or adjust its trajectory. The bubble's motion is determined entirely by its initial configuration, with no possibility of mid-flight modification.

3.2 Superluminal Travel as Ballistic Propagation

Because of this causal isolation, superluminal motion behaves like ballistic propagation. The vehicle becomes a faster-than-light bullet, moving blindly through spacetime. Any deviation from the intended trajectory—whether due to external gravitational fields, density gradients, or spacetime curvature—cannot be detected or compensated for. The bubble cannot be steered, stabilised, or navigated, and its endpoint cannot be guaranteed.

This ballistic behaviour is not merely impractical; it is structurally unavoidable. Any mechanism that permits superluminal motion must, by definition, outrun the propagation of information in the surrounding spacetime. As a result, the vehicle cannot receive external updates about its environment, nor can it send internal updates outward. Navigation becomes impossible in principle.

3.3 Implications for Physical Realisability

Within the Triadic Cosmos framework, ballistic superluminal propagation is incompatible with the requirements for closure-preserving physical systems. A navigable vehicle must maintain continuous causal coupling with its environment, allowing for feedback, correction, and stabilisation. Superluminal bubbles violate this requirement by severing all causal links between interior and exterior closures.

Thus, even if superluminal travel were technically feasible, it would fail to satisfy the operational conditions necessary for controlled motion. The unguided projectile argument therefore forms the second pillar of the broader no-go theorem: superluminal travel cannot be navigated, cannot be stabilised, and cannot be made safe or functional within any causally coherent physical ontology.

4 Technical Impossibility: Exotic Requirements and Triadic Filtering

Even if empirical evidence and navigational constraints are set aside, superluminal travel remains technically unrealizable within any physically coherent framework. All known proposals for faster-than-light transport rely on mechanisms that require forms of matter, energy, or spacetime manipulation that violate established physical principles. Within the Triadic Cosmos ecosystem, these mechanisms are further excluded by triadic filtering rules and local-closure constraints, which prohibit any structure that breaks causal embedding or boundary normalisation. This section examines the technical barriers that render superluminal transport physically impossible.

4.1 Dependence on Exotic Matter and Negative-Energy Configurations

Warp bubbles, wormholes, and related constructs require stress-energy tensors that cannot be produced by any known or theoretically permissible form of matter. Alcubierre-type warp metrics demand regions of negative energy density, violating classical energy conditions and requiring exotic matter that has no empirical or theoretical support within the triadic ontology. Wormholes and Krasnikov tubes similarly rely on non-physical configurations of spacetime curvature that cannot be generated or sustained by any triadically admissible physical process.

Within the Triadic Cosmos, triadic filtering explicitly excludes any entity or mechanism that requires closure-breaking physics. Exotic matter, negative-energy fields, and non-unitary spacetime configurations cannot be embedded within the triadic closure hierarchy and therefore cannot exist as physically realizable structures.

4.2 Incompatibility with Local Closure and Triadic Embedding

Superluminal mechanisms also violate the structural requirement of local closure. A physically admissible system must maintain consistent causal embedding within its parent closure, ensuring that all interactions and boundary conditions remain compatible with the surrounding spacetime. Warp bubbles and wormholes necessarily create regions of spacetime that cannot be embedded within their parent closures, producing child closures that violate the Universal Closure–Boundary Normalisation Principle.

This incompatibility is not a minor technical obstacle but a fundamental structural failure. Any mechanism that produces a closure-breaking region cannot be realized within the Triadic Cosmos, regardless of the availability of energy, technology, or engineering sophistication.

4.3 Exclusion of Alternative Superluminal Mechanisms

Beyond warp bubbles and wormholes, other proposed superluminal mechanisms—tachyonic channels, quantum tunneling shortcuts, and higher-dimensional bypasses—are likewise excluded. Tachyonic propagation violates unitarity and produces non-physical dispersion relations. Quantum tunneling cannot transmit information superluminally and cannot be scaled to macroscopic transport. Higher-dimensional shortcuts require non-triadic ontologies that break closure consistency and boundary normalisation.

Triadic filtering systematically removes all such mechanisms from the admissible physical ontology. Any proposal that relies on non-local, non-unitary, or closure-breaking physics is rejected at the structural level, independent of empirical or causal considerations.

4.4 Technical Impossibility as a Structural Constraint

The technical impossibility of superluminal travel is therefore not a matter of engineering difficulty or insufficient technological development. It is a structural constraint arising from the requirements of closure, boundary normalisation, and triadic embedding. No physically coherent mechanism exists that can produce or sustain superluminal motion without violating these foundational principles.

Technical impossibility thus forms the third pillar of the broader no-go theorem: even in the absence of empirical and causal objections, superluminal travel cannot be realized within any triadically consistent physical framework.

5 Causality Violations: Holographic, Electromagnetic, Gravitational, and Temporal Breakdown

Beyond empirical absence and technical impossibility, superluminal travel fails at the deepest structural level: it violates causality in multiple independent ways. These violations are not optional side effects but unavoidable consequences of faster-than-light propagation. Any mechanism that outruns the causal structure of spacetime necessarily disrupts holographic consistency, electromagnetic information flow, gravitational-wave propagation, and classical cause–effect ordering. Within the Triadic Cosmos framework, each of these failures independently breaks the Universal Closure–Boundary Normalisation Principle (UCBNP), and together they demonstrate that superluminal travel is incompatible with any coherent physical ontology.

5.1 Holographic Causality Breakdown

In holographic frameworks, every bulk event must correspond to a boundary representation, and causal relationships in the bulk must be faithfully encoded in the boundary. Superluminal motion disrupts this mapping by creating bulk regions that cannot be reconstructed from boundary data. The interior of a superluminal bubble becomes a causally isolated child closure that cannot embed within its parent closure, violating entanglement-wedge nesting and breaking bulk–boundary unitarity.

This failure is structural: holographic consistency requires that no region of spacetime outrun the propagation of boundary information. Superluminal bubbles violate this requirement by creating zones that cannot be holographically normalised, thereby breaking UCBNP at the foundational level.

5.2 Electromagnetic Information Propagation Failure

Electromagnetic signals propagate at the speed of light. Any object traveling faster than light necessarily outruns all EM-based information channels. This produces several unavoidable consequences:

- external EM signals cannot reach the bubble interior,
- internal EM signals cannot escape the bubble,
- no communication or sensing is possible across the bubble boundary,
- EM fields become compressed into superluminosity zones ahead of the bubble.

These effects break microcausality in quantum field theory and violate the requirement that information propagation remain consistent with local closure. The bubble becomes a region where EM causality is fundamentally disrupted.

5.3 Gravitational-Wave Latency and Curvature Update Failure

Gravitational waves propagate at the speed of light. A superluminal bubble therefore outruns the propagation of curvature updates in the surrounding spacetime. The bubble moves into regions whose geometry has not yet been updated to reflect its presence, producing a mismatch between the bubble's stress-energy configuration and the curvature information available to the external universe.

This latency produces:

- delayed gravitational-wave trails,
- curvature discontinuities,
- violations of the Einstein field equations' local consistency,
- breakdown of causal closure in the gravitational sector.

Within the Triadic Cosmos, such gravitational latency constitutes a direct violation of closure embedding and boundary normalisation, making superluminal motion incompatible with triadic gravitational architecture.

5.4 Classical Cause–Effect Ordering and Temporal Paradoxes

Superluminal motion also breaks classical causality. Because electromagnetic signals cannot propagate faster than light, any faster-than-light communication requires a physical messenger device that itself travels superluminally. This makes the causal paradox particularly clear when considering a single spacecraft capable of superluminal travel.

A simple paradox arises as follows:

- A spacecraft departs from location A and travels superluminally to location B.
- Upon arrival at B, it transmits an ordinary electromagnetic message back to A describing what it has observed.
- After a short delay, the spacecraft departs B and returns to A using superluminal travel.

- The spacecraft arrives back at A *before* the electromagnetic message it transmitted from B can reach A.
- The spacecraft can then receive its own message at A, even though from A's perspective the spacecraft has not yet arrived at B at the time the message was sent.

In this sequence, the causal order of events is inverted: the spacecraft returns before its own electromagnetic transmission arrives. This produces a closed causal loop in which the spacecraft receives information about an event that, according to A's local timeline, has not yet occurred. The result is retrocausal signalling, violations of Lorentz-invariant cause–effect ordering, and the formation of closed timelike curves.

Such paradoxes are incompatible with any physical ontology that preserves temporal coherence, including the triadic closure hierarchy. Temporal causality cannot be maintained in the presence of superluminal transport.

5.5 Causality Violations as Structural Failure

Each of these causal breakdowns—holographic, electromagnetic, gravitational, and temporal—independently violates the Universal Closure–Boundary Normalisation Principle. Together, they demonstrate that superluminal travel is not merely problematic but structurally impossible. No universe that maintains coherent causal propagation, holographic completeness, and closure consistency can support faster-than-light transport.

Causality violations thus form the fourth pillar of the broader no-go theorem: superluminal travel breaks the causal fabric of spacetime at every level, making it incompatible with any triadically consistent physical framework.

6 Unified No-Go Theorem: Structural Impossibility of Superluminal Travel

The preceding sections have examined superluminal travel from four independent perspectives: empirical constraints, navigational limitations, technical impossibility, and causal breakdown. Each perspective provides a self-contained argument against the physical realizability of faster-than-light transport. In this section, we integrate these arguments into a unified no-go theorem demonstrating that superluminal travel is structurally impossible within any universe that maintains coherent closure, boundary normalisation, and causal propagation. The result is not merely a collection of objections but a comprehensive proof of incompatibility.

6.1 Synthesis of Empirical, Technical, and Causal Failures

The empirical analysis (Section A) shows that superluminal transport would produce unmistakable observational signatures, including electromagnetic compression artefacts, gravitational-wave latency trails, and superluminosity zones. The absence of such phenomena across all astronomical surveys constitutes strong evidence that no superluminal mechanisms operate at any significant scale.

The navigational analysis (Section B) demonstrates that even if superluminal transport were technically possible, it would behave as an unguided projectile. The causal horizon surrounding any superluminal bubble prevents information exchange with the external universe, eliminating feedback, sensing, routing, and stabilisation. Superluminal motion is therefore operationally unusable.

The technical analysis (Section C) shows that all known superluminal mechanisms require exotic matter, negative-energy configurations, or closure-breaking physics that violate triadic filtering and local closure constraints. These mechanisms cannot be embedded within a triadically consistent ontology and cannot be physically realized.

The causal analysis (Section D) demonstrates that superluminal motion violates causality in four independent ways: holographic bulk–boundary consistency, electromagnetic information propagation,

gravitational-wave update latency, and classical cause–effect ordering. Each violation independently breaks the Universal Closure–Boundary Normalisation Principle.

6.2 Formal Statement of the No-Go Theorem

Theorem (Unified Superluminal No-Go Theorem). *In any universe that satisfies closure consistency, boundary normalisation, holographic completeness, and causal propagation at finite speed, superluminal travel is structurally impossible. No mechanism can produce or sustain faster-than-light motion without violating at least one of the following:*

1. *empirical consistency with electromagnetic and gravitational backgrounds,*
2. *operational requirements for navigable and feedback-driven motion,*
3. *technical constraints on admissible stress-energy configurations,*
4. *causal coherence across holographic, electromagnetic, gravitational, and temporal domains.*

Any violation of these principles produces closure-breaking regions that cannot be embedded within their parent closures, thereby violating the Universal Closure–Boundary Normalisation Principle and rendering the mechanism physically inadmissible.

6.3 Implications for the Triadic Cosmos and Beyond

Within the Triadic Cosmos framework, the theorem confirms that superluminal travel is excluded at the structural level. The ecosystem’s predictive architecture asserts that any closure-breaking phenomenon must produce observable distortions across electromagnetic, gravitational, and causal layers. The absence of such distortions, combined with the technical and causal failures outlined above, demonstrates that superluminal transport cannot exist within the triadic ontology.

More broadly, the theorem applies to any universe that maintains coherent causal propagation and holographic completeness. The impossibility of superluminal travel is therefore not a peculiarity of the Triadic Cosmos but a universal constraint arising from the fundamental structure of spacetime.

6.4 Conclusion of the No-Go Argument

By integrating empirical non-realisation, navigational impossibility, technical exclusion, and causal breakdown, we obtain a comprehensive and accessible refutation of superluminal travel. The unified no-go theorem shows that faster-than-light transport is not merely technologically challenging or currently unattainable; it is structurally incompatible with any physically coherent universe.

Superluminal travel cannot exist. Not as technology, not as geometry, and not as a causal phenomenon.

7 Discussion

The analysis presented in this paper demonstrates that superluminal travel fails on empirical, operational, technical, and causal grounds. Although each pillar independently excludes faster-than-light transport, their combined force reveals a deeper structural incompatibility. The Triadic Cosmos framework does not merely lack mechanisms for superluminal motion; it actively prohibits them through closure constraints, boundary normalisation, and causal propagation rules. This discussion section reflects on the broader implications of these findings and situates them within the conceptual ecosystem of the Triadic Cosmos.

7.1 The Dyson Spaceship as an Extreme Thought Experiment

The Dyson Spaceship scenario introduced in Section A serves as a deliberately exaggerated thought experiment. By imagining a stellar-scale propulsion system enclosed within a Dyson sphere and driving a superluminal warp bubble, we explore the upper limits of what an advanced Kardashev Type II civilization might attempt. This scenario is intentionally extreme: if superluminal travel were possible at any scale, a stellar engine would represent the most powerful and least resource-constrained implementation.

The value of this thought experiment lies not in its plausibility but in its diagnostic power. A warp bubble driven by stellar output would produce overwhelming electromagnetic compression, gravitational-wave latency trails, and spacetime shear artefacts. These signatures would be visible across cosmological distances and would be impossible to conceal. Their absence in all observational data strongly suggests that no such superluminal structures exist, even among hypothetical high-level civilizations. The Dyson Spaceship thus functions as a conceptual stress test: if superluminal travel were real, the universe would reveal it.

7.2 From Thought Experiment to Structural Exclusion

The flow of the argument across Sections A–D shows that the Dyson Spaceship is not an isolated case but a representative example of a broader pattern. Smaller warp bubbles, wormholes, tachyonic channels, and higher-dimensional shortcuts would all produce analogous distortions, albeit at reduced scale. Their absence in observational data reinforces the empirical pillar of the no-go theorem.

The unguided projectile argument (Section B) further demonstrates that even a technically feasible warp bubble would be operationally unusable. The causal isolation of the bubble interior eliminates feedback, navigation, and stabilisation, reducing the vehicle to blind ballistic propagation. This operational failure is independent of the scale of the mechanism and applies equally to megastructures and microscopic bubbles.

Technical constraints (Section C) show that all known superluminal mechanisms require exotic matter, negative-energy configurations, or closure-breaking physics that cannot be embedded within the triadic ontology. Triadic filtering systematically excludes these mechanisms, confirming earlier predictions within the ecosystem regarding the impossibility of warp drives and related constructs.

Finally, the causal analysis (Section D) reveals that superluminal motion violates holographic consistency, electromagnetic information propagation, gravitational-wave update coherence, and classical cause–effect ordering. These violations independently break the Universal Closure–Boundary Normalisation Principle, confirming that superluminal travel is incompatible with any causally coherent universe.

7.3 Confirmation of Prior Predictions

The unified no-go theorem formulated in Section E confirms and strengthens earlier predictions within the Triadic Cosmos ecosystem. Prior analyses of local closure, triadic filtering, and boundary normalisation suggested that warp drives, wormholes, and other superluminal constructs could not be physically realized. The present work provides a comprehensive and accessible synthesis of these predictions, demonstrating that superluminal travel is excluded not only by triadic principles but by the fundamental structure of spacetime itself.

7.4 Broader Implications

The impossibility of superluminal travel has significant implications for cosmology, astrophysics, and speculative engineering. It suggests that all physically admissible forms of transport must operate within the causal structure of spacetime and that any attempt to circumvent this structure leads to closure-breaking regions that cannot be embedded within a coherent ontology. The Triadic Cosmos thus aligns with and extends the constraints of classical relativity, holography, and quantum field theory.

In summary, the Dyson Spaceship thought experiment highlights the empirical stakes of superluminal travel, while the subsequent analysis reveals its deeper structural impossibility. The unified no-go

theorem demonstrates that faster-than-light transport cannot exist as technology, geometry, or causal phenomenon. The Triadic Cosmos framework not only predicts this outcome but provides the conceptual tools to understand why it must be so.

8 Conclusion

This paper has developed a comprehensive and multi-layered refutation of superluminal travel within the Triadic Cosmos framework. By examining empirical constraints, operational limitations, technical barriers, and causal breakdowns, we have shown that faster-than-light transport is excluded at every level of physical coherence. The argument does not rely on any single failure mode; rather, it emerges from the convergence of independent structural constraints that collectively render superluminal motion impossible.

The Dyson Spaceship thought experiment served as a conceptual upper bound, illustrating the extreme observational consequences that a stellar-scale superluminal propulsion system would produce. Its absence in all astronomical data reinforces the empirical pillar of the no-go theorem. Smaller-scale warp bubbles and related constructs would likewise generate detectable electromagnetic and gravitational artefacts, none of which have ever been observed.

Operational analysis revealed that even a technically feasible warp bubble would be unusable. The causal isolation of the bubble interior eliminates feedback, sensing, routing, and stabilisation, reducing any superluminal vehicle to blind ballistic propagation. This unguided projectile behaviour is incompatible with the requirements for navigable motion in any coherent physical ontology.

Technical analysis demonstrated that all known superluminal mechanisms require exotic matter, negative-energy configurations, or closure-breaking physics that violate triadic filtering and local closure constraints. These mechanisms cannot be embedded within the triadic ontology and cannot be physically realized.

Causal analysis showed that superluminal motion violates holographic consistency, electromagnetic information propagation, gravitational-wave update coherence, and classical cause–effect ordering. Each violation independently breaks the Universal Closure–Boundary Normalisation Principle, confirming that superluminal transport is incompatible with any universe that maintains coherent causal structure.

Together, these findings form the unified no-go theorem presented in Section E: superluminal travel is structurally impossible not only within the Triadic Cosmos but within any causally consistent universe. The Triadic Cosmos framework not only predicts this outcome but provides the conceptual tools to understand why it must be so. The impossibility of faster-than-light transport is therefore not a technological limitation or an engineering challenge; it is a fundamental consequence of the structure of spacetime, closure, and causality.

Superluminal travel cannot exist. Not as technology, not as geometry, and not as a causal phenomenon. The Triadic Cosmos confirms this conclusion with clarity, coherence, and structural inevitability.

Epilogue

Superluminal travel has inspired decades of speculation, engineering fantasies, and cosmological imaginings. Yet when examined through the combined lenses of empirical constraints, operational limitations, technical admissibility, and causal coherence, the concept dissolves under its own impossibility. The Triadic Cosmos framework reveals that faster-than-light motion is not merely unrealised, but structurally forbidden.

And so, the superluminal dream reaches its end.

A Related Classical Work and External Constraints

Although the Triadic Cosmos framework provides a unique structural basis for the unified no-go theorem developed in this paper, it is instructive to situate the argument within the broader landscape of classical and modern physics. No existing work formulates a superluminal exclusion principle in terms of triadic closure, boundary normalisation, or holographic causal embedding. Nevertheless, several families of external results align with and reinforce the conclusions reached here. This appendix briefly surveys these related works and highlights their conceptual connections.

A.1 Classical Relativistic No-Go Results

A substantial body of literature demonstrates that superluminal transport is incompatible with general relativity unless exotic matter or negative-energy configurations are introduced. Alcubierre’s original warp metric [6] requires stress-energy tensors that violate classical energy conditions, a point further developed by Hiscock [7] and by Pfenning and Ford [8], who showed that quantum inequalities forbid the sustained negative-energy densities required for warp bubbles and wormholes. Visser’s work on Lorentzian wormholes [9] similarly demonstrates that traversable wormholes demand non-physical matter distributions. These results align with the Triadic Cosmos conclusion that exotic matter and closure-breaking physics cannot be embedded within a consistent ontology.

A.2 Holographic and Bulk–Boundary Causality Constraints

Modern holographic frameworks impose strict requirements on causal propagation between bulk and boundary degrees of freedom. The Ryu–Takayanagi formula [10] and subsequent developments in entanglement-wedge reconstruction [11, 12] show that bulk regions must be reconstructible from boundary data. Superluminal bubbles violate this requirement by creating causally isolated child closures that cannot be holographically normalised. Although these works do not address superluminal travel directly, they reinforce the Triadic Cosmos argument that holographic consistency forbids faster-than-light propagation.

A.3 Gravitational-Wave Propagation and Curvature Update Coherence

The propagation of gravitational waves at the speed of light, first derived by Einstein and confirmed observationally by LIGO/Virgo [13], imposes strict constraints on curvature update coherence. Any object moving faster than light necessarily outruns the propagation of curvature information, producing gravitational-wave latency trails and curvature discontinuities. While classical general relativity does not frame this as a closure violation, the Triadic Cosmos interpretation shows that such latency breaks boundary normalisation and causal embedding.

A.4 Classical Causality Paradoxes and Temporal Coherence

Classical analyses of superluminal signalling, including Tolman’s early paradox [14], Recami’s tachyon studies [15], and modern treatments of causal loops [16], demonstrate that faster-than-light motion leads to retrocausal signalling and closed timelike curves. These paradoxes align with the Triadic Cosmos conclusion that temporal coherence cannot be preserved in the presence of superluminal transport. The spacecraft paradox formulated in Section D provides a triadically consistent version of these classical results.

A.5 Synthesis

None of the works surveyed here provide a unified structural exclusion principle for superluminal travel. However, each contributes a piece of the broader picture: exotic matter is forbidden, holographic consistency is disrupted, gravitational-wave propagation is violated, and temporal coherence collapses. The

Triadic Cosmos framework integrates these disparate constraints into a single structural no-go theorem, demonstrating that superluminal travel is impossible not only within triadic ontology but within any causally coherent universe.

B Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

<https://github.com/triadic-cosmos>

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